

A Separable Framework for Joint Radial Velocity, Template Weight, Continuum, and Spectral Resolution Fitting of Stellar Spectra

Abstract

We derive the spectral-fitting framework implemented in this project, which simultaneously recovers a radial velocity v , the weights $\mathbf{w}$ of a learned (PCA) template basis, a smooth Legendre continuum $\mathbf{c}$, and optionally the instrumental spectral resolution R , from a single observed stellar spectrum. The key structural fact, exploited throughout, is *separability*: working in log-wavelength $x = \ln \lambda$ and in log-flux, the radial velocity becomes a uniform translation, the resolution becomes a fixed-width Gaussian, and the continuum becomes additive, so that at fixed values of the nonlinear parameters (v, R) the model is *linear* in $\boldsymbol{\theta} = (\mathbf{w}, \mathbf{c})$ and admits an exact weighted least-squares solution. We build the model in seven steps: the linear template fit; the radial-velocity translation and per-shift solve (the cross-correlation of [Tonry & Davis 1979](#), generalised to a template basis); FFT acceleration of the entire velocity scan; the convexity structure of the joint problem (convex in the linear block, multimodal in v); the additive Legendre continuum (in the spirit of the multiplicative/additive polynomials of [pPXF](#), [Cappellari 2017](#)); the on-the-fly Gaussian broadening for resolution, together with an honest accounting of the log-space convolution approximation; and finally the actual optimiser — a coarse Fourier scan to locate the global basin followed by a JAX-autodiff damped-Newton refinement of the profiled χ^2 , with covariances from its Hessian. The PCA-template basis and the external resolution/redshift scan parallel the classification approach of [Bolton et al. \(2012\)](#).

Notation

All symbols used in this document are collected in Table 1 and are not redefined elsewhere. Vectors are bold lowercase (the log-flux data $\mathbf{Y}$ is the lone bold-uppercase vector), matrices bold uppercase; the per-pixel index is $i = 1, \dots, n$; a hat denotes an estimate.

1 Setup and the linear template fit

We begin by resampling both the observed spectrum and the templates onto a single uniform grid in log-wavelength, $x_i = x_1 + (i - 1)\delta$ with $i = 1, \dots, n$ and constant spacing δ . Two facts motivate this grid: a Doppler shift is a uniform translation in x (Section 2), and at constant resolving power the line-spread function (LSF) has a constant width in x (Section 6).

We model the spectrum in *log-flux*, $Y_i = \ln d_i$. This is the natural space for two reasons. First, the observed flux is the product of a smooth continuum and a normalised (rectified) spectrum, and the logarithm turns this product into a sum, making the continuum an *additive* term (Section 5). Second, the template basis is built in log-rectified flux, as follows. Each model spectrum is converted to log-flux, a robust low-order Legendre fit to the upper (continuum) envelope is subtracted to *rectify* it, and a PCA is performed on the rectified spectra. This yields a mean rectified log-flux spectrum

$\mu(x)$ and orthonormal principal components $\phi_k(x)$, $k = 1, \dots, K$, that capture the *line* information. We collect them as the columns of the template block,

$$\mathbf{T} = [\mu \mid \phi_1 \mid \dots \mid \phi_K], \quad T_0 \equiv \mu, \quad T_k \equiv \phi_k, \quad (1)$$

so that $\mathbf{T}$ has $K + 1$ columns. The mean μ is treated as a free-weight template (column T_0 , weight w_0), so it is Doppler-shifted like any component.

The simplest fitting problem models the observed log-rectified flux as a linear combination of the templates,

$$Y(x) = \sum_{k=0}^K w_k T_k(x), \quad (2)$$

with weight vector $\mathbf{w} = (w_0, \dots, w_K)$.

Data covariance. The noise is naturally specified on the linear flux d_i with dispersion σ_i . Propagating to log-flux via $Y_i = \ln d_i$ and the first-order (delta-method) linearisation gives the variance of Y_i ,

$$\sigma_{Y,i} \approx \left| \frac{\partial \ln d}{\partial d} \right| \sigma_i = \frac{\sigma_i}{d_i}, \quad \implies \quad C_i = \sigma_{Y,i}^2 = \left(\frac{\sigma_i}{d_i} \right)^2, \quad (3)$$

so the inverse-variance weight is $C_i^{-1} = (d_i/\sigma_i)^2$. Bad or out-of-range pixels (non-finite d_i or σ_i , $d_i \leq 0$, masked pixels) are excised by setting $C_i^{-1} = 0$ (equivalently $C_i \rightarrow \infty$); we write $\mathbf{C} = \text{diag}(C_i)$ and let $n_{\text{good}} = \#\{i : C_i^{-1} > 0\}$ be the number of usable pixels.

Weighted least squares. With the design matrix taken to be the template block $\mathbf{T}$, the objective is the weighted residual sum of squares,

$$\chi^2(\mathbf{w}) = (\mathbf{Y} - \mathbf{T}\mathbf{w})^\top \mathbf{C}^{-1} (\mathbf{Y} - \mathbf{T}\mathbf{w}). \quad (4)$$

This is a quadratic in $\mathbf{w}$; setting $\nabla_{\mathbf{w}} \chi^2 = 0$ yields the normal equations

$$\mathbf{N}\mathbf{w} = \mathbf{r}, \quad \mathbf{N} = \mathbf{T}^\top \mathbf{C}^{-1} \mathbf{T}, \quad \mathbf{r} = \mathbf{T}^\top \mathbf{C}^{-1} \mathbf{Y}, \quad (5)$$

with the unique solution and minimum value

$$\hat{\mathbf{w}} = \mathbf{N}^{-1} \mathbf{r}, \quad \chi^2(\hat{\mathbf{w}}) = \mathbf{Y}^\top \mathbf{C}^{-1} \mathbf{Y} - \mathbf{r}^\top \mathbf{N}^{-1} \mathbf{r}. \quad (6)$$

The Hessian of (4) is $2\mathbf{N} = 2\mathbf{T}^\top \mathbf{C}^{-1} \mathbf{T} \succeq 0$, so the problem is a convex quadratic; it is *strictly* convex — with a unique minimiser — if and only if $\mathbf{T}^\top \mathbf{C}^{-1} \mathbf{T}$ is positive definite, which requires the weighted template columns to be linearly independent. If they are not (a (near-)singular $\mathbf{N}$, e.g. from collinear components or too few good pixels), the solve is regularised: a Tikhonov ridge $\mathbf{N} \rightarrow \mathbf{N} + \rho \mathbf{I}$, a truncated SVD, or a non-negativity constraint (NNLS) when physical positivity of the weights is desired.

2 Radial velocity as translation: the per-shift linear solve

A radial velocity v Doppler-shifts every wavelength by the factor $1 + v/c$ (non-relativistically). In log-wavelength this is a *uniform translation*,

$$\lambda \rightarrow \lambda(1 + v/c) \iff x \rightarrow x + \Delta x, \quad \Delta x \equiv \ln\left(1 + \frac{v}{c}\right), \quad (7)$$

which on the pixel grid is a shift of $p = \Delta x / \delta$ pixels. The model templates therefore become $T_k(x) \rightarrow T_k(x - \Delta x)$, i.e. each column of $\mathbf{T}$ is translated by p pixels. We write the shifted design as $\mathbf{T}(p)$, with columns $[T_k(x_i - p\delta)]_i$.

The forward model (2) now reads $Y(x) = \sum_k w_k T_k(x - \Delta x)$. Crucially, it is *linear in $\mathbf{w}$ only at a fixed shift p* ; the dependence on p is nonlinear. This structure is *separable*: we scan the single nonlinear parameter p on the *outside* and solve the linear weight problem exactly on the *inside*. For each trial shift,

$$\mathbf{N}(p) = \mathbf{T}(p)^\top \mathbf{C}^{-1} \mathbf{T}(p), \quad \mathbf{r}(p) = \mathbf{T}(p)^\top \mathbf{C}^{-1} \mathbf{Y}, \quad \hat{\mathbf{w}}(p) = \mathbf{N}(p)^{-1} \mathbf{r}(p), \quad (8)$$

and the *profiled* objective, obtained by minimising out $\mathbf{w}$ as in (6), is

$$\chi^2(p) = \min_{\mathbf{w}} \chi^2(\mathbf{w}, p) = \mathbf{Y}^\top \mathbf{C}^{-1} \mathbf{Y} - \mathbf{r}(p)^\top \mathbf{N}(p)^{-1} \mathbf{r}(p). \quad (9)$$

The radial-velocity estimate is the global minimiser,

$$\hat{p} = \arg \min_p \chi^2(p), \quad \hat{v} = c(e^{\hat{p}\delta} - 1). \quad (10)$$

For a single template ($K = 0$, no continuum, uniform weights) the cross term $\mathbf{r}(p)^\top \mathbf{N}(p)^{-1} \mathbf{r}(p)$ reduces to the squared template–data cross-correlation, so maximising it over p is exactly the classical cross-correlation velocity measurement of [Tonry & Davis \(1979\)](#). Equation (9) is its generalisation to a multi-column template *basis* (and, in Sections 5–6, to a joint continuum and resolution model).

3 FFT acceleration of the velocity scan

A direct evaluation of (9) on a grid of n_{shifts} trial velocities, each requiring $\mathbf{T}(p)^\top \mathbf{C}^{-1} \mathbf{T}(p)$ and $\mathbf{T}(p)^\top \mathbf{C}^{-1} \mathbf{Y}$, costs $O(n_{\text{shifts}} n K^2)$. The expensive shift-dependent pieces are, however, all *cross-correlations of fixed vectors against a shifted template*, which a single FFT per template column delivers at *all* shifts simultaneously.

Consider the right-hand side. Its k -th template entry, as a function of the integer lag p , is

$$b_{T,k}(p) = \sum_i T_k(x_i - p\delta) C_i^{-1} Y_i = [T_k \star (\mathbf{C}^{-1} \mathbf{Y})](p), \quad (11)$$

a discrete cross-correlation of the fixed weighted-data vector $\mathbf{C}^{-1} \mathbf{Y}$ against T_k . By the cross-correlation theorem, for any two real vectors t, s of length n ,

$$\text{corr}(t, s) = \text{irfft}(\overline{\text{rfft}(t)} \cdot \text{rfft}(s)), \quad (12)$$

where rfft/irfft are the real forward/inverse FFTs, the product is elementwise over frequencies ω , and the overline denotes complex conjugation. Hence

$$\mathbf{b}_T(\cdot) = \text{irfft}(\overline{\text{rfft}(\mathbf{T})} \cdot \text{rfft}(\mathbf{C}^{-1} \mathbf{Y})), \quad (13)$$

computed once for every lag and every template column. A fractional (sub-pixel) shift is applied in the same domain as a linear phase ramp,

$$\text{rfft}[T_k(\cdot - p\delta)] = \text{rfft}(T_k) \cdot e^{-2\pi i \omega p}, \quad (14)$$

so the model template at any real-valued p is recovered by $\text{irfft}(\text{rfft}(T_k) e^{-2\pi i \omega p})$.

The normal-matrix block $\mathbf{M}_{TT}(p) = \mathbf{T}(p)^\top \mathbf{C}^{-1} \mathbf{T}(p)$ depends on p *only* through the (slowly varying) weights $\mathbf{C}^{-1}$: if $\mathbf{C}^{-1}$ were constant, $\mathbf{M}_{TT}$ would be exactly shift-invariant. We therefore approximate $\mathbf{M}_{TT}(p) \approx \mathbf{M}_{TT}(0)$ during the scan (cheap, evaluated once) and recompute $\mathbf{M}_{TT}$ *exactly* at the refined optimum (Section 8).

Circular-edge handling. The FFT correlation in (12) is *circular*: content shifted off one end wraps to the other. To make the circular correlation equal the desired linear one over the region of interest, we zero the data weights C_i^{-1} in a band of width equal to the maximum lag $|p|_{\max}$ at *both* ends of the grid. Within the interior valid region, no wrapped pixel ever carries weight for any evaluated shift, so the circular and linear correlations coincide.

Cost. The whole velocity scan now costs $O(K n \log n)$ — a handful of FFTs of length n , one per template column — instead of $O(n_{\text{shifts}} n K^2)$, a decisive saving when n_{shifts} and n are both large.

4 Joint RV and weights: the convexity structure

A natural question is whether the joint problem in $(\mathbf{w}, p)$ is convex. The answer is block-structured, and worth stating precisely because it dictates the optimisation strategy.

Convex in the linear block. At any fixed shift p , the objective $\chi^2(\mathbf{w}, p)$ in (4) (with $\mathbf{T} \rightarrow \mathbf{T}(p)$) is a convex quadratic in $\mathbf{w}$ with Hessian $2\mathbf{N}(p) \succeq 0$; it is globally and (when $\mathbf{N}(p) \succ 0$) uniquely solvable in closed form by (8). There are no spurious local minima in the weights.

Non-convex in the nonlinear parameter. The profiled objective $\chi^2(p)$ of (9), obtained by minimising out $\mathbf{w}$, is *not* convex in p . Up to the constant $\mathbf{Y}^T \mathbf{C}^{-1} \mathbf{Y}$, it is the (negated) generalised cross-correlation surface $\mathbf{r}(p)^T \mathbf{N}(p)^{-1} \mathbf{r}(p)$, which is inherently *multimodal*: the quasi-periodic absorption-line pattern and the self-similarity of the templates produce sidelobes (secondary minima) at lags offset from the truth by characteristic line spacings, with one dominant minimum at the true velocity.

Consequence. The joint problem is thus *convex in the linear block, non-convex (multimodal) in the nonlinear parameter*. Globally it is not convex; block-wise (in $\mathbf{w}$) and locally (near the dominant peak) it is. This is exactly why the method uses a *global coarse scan* over p to identify the basin of the dominant minimum (Section 3), followed by a *local refinement* (Section 8) in which the main peak is smooth and approximately parabolic, hence locally well-behaved (locally convex). The short answer to “is it convex?” is: *no globally, yes block-wise and locally*.

5 Adding the continuum within the same linear model

The observed spectrum is the product of a smooth continuum and the rectified (line) spectrum. In log-flux this product becomes a sum, so a smooth continuum is an *additive* term that we represent with low-order Legendre polynomials P_m , $m = 0, \dots, L$, evaluated on x mapped affinely onto $[-1, 1]$ across the fit range,

$$\tilde{x}_i = 2 \frac{x_i - x_1}{x_n - x_1} - 1 \in [-1, 1], \quad (\mathbf{P})_{im} = P_m(\tilde{x}_i). \quad (15)$$

The full forward model in log-flux is then

$$Y(x) = \underbrace{\sum_{k=0}^K w_k T_k(x - \Delta x)}_{\text{rectified line basis}} + \underbrace{\sum_{m=0}^L c_m P_m(x)}_{\text{log-continuum}}. \quad (16)$$

We augment the design matrix by appending the continuum block,

$$\mathbf{A}(p) = [\mathbf{T}(p) \mid \mathbf{P}], \quad \boldsymbol{\theta} = (\mathbf{w}, \mathbf{c}), \quad (17)$$

with $\mathbf{A}$ of size $n \times D$, $D = (K+1) + (L+1)$. The continuum block is *shift-independent*, so its contributions to the normal equations are precomputed once,

$$\mathbf{M}_{PP} = \mathbf{P}^\top \mathbf{C}^{-1} \mathbf{P}, \quad \mathbf{b}_P = \mathbf{P}^\top \mathbf{C}^{-1} \mathbf{Y} \quad (\text{independent of } p). \quad (18)$$

Only the template auto-block and the template–continuum cross-block depend on p :

$$\mathbf{M}_{TT}(p) = \mathbf{T}(p)^\top \mathbf{C}^{-1} \mathbf{T}(p), \quad \mathbf{M}_{TP}(p) = \mathbf{T}(p)^\top \mathbf{C}^{-1} \mathbf{P}, \quad (19)$$

and the cross-block $\mathbf{M}_{TP}(p)$ is, like (11), a cross-correlation of the fixed weighted-continuum columns $\mathbf{C}^{-1} \mathbf{P}$ against the shifted templates — so it too is obtained at all lags by FFT, $\mathbf{M}_{TP}(\cdot) = \text{irfft}(\overline{\text{rfft}(\mathbf{T})} \cdot \text{rfft}(\mathbf{C}^{-1} \mathbf{P}))$. The block normal equations are

$$\mathbf{N}(p) = \begin{bmatrix} \mathbf{M}_{TT}(p) & \mathbf{M}_{TP}(p) \\ \mathbf{M}_{TP}(p)^\top & \mathbf{M}_{PP} \end{bmatrix}, \quad \mathbf{r}(p) = \begin{bmatrix} \mathbf{b}_T(p) \\ \mathbf{b}_P \end{bmatrix}, \quad (20)$$

with the inner solve and profiled objective exactly as before,

$$\hat{\boldsymbol{\theta}}(p) = \mathbf{N}(p)^{-1} \mathbf{r}(p), \quad \chi^2(p) = \mathbf{Y}^\top \mathbf{C}^{-1} \mathbf{Y} - \mathbf{r}(p)^\top \mathbf{N}(p)^{-1} \mathbf{r}(p). \quad (21)$$

Why separate the continuum from the templates. Because the templates are *rectified* (their pseudo-continuum has been removed before the PCA), they carry essentially only line information and are approximately zero in the continuum. The smooth flux level is carried entirely by the orthogonal Legendre block. This separation of roles is what avoids the classic *template/continuum degeneracy*: were the templates allowed to carry broadband shape, a low-order continuum could partially mimic and absorb it, leaving $\mathbf{N}$ ill-conditioned and $(\mathbf{w}, \mathbf{c})$ poorly determined. Rectifying the templates and modelling the continuum as a separate, smooth, additive block keeps the two contributions linearly independent (to the extent that line structure and low-order polynomials are dissimilar) and the solve well-conditioned. This is the same design principle as the additive and multiplicative polynomial continua in pPXF (Cappellari, 2017) and the PCA-template plus polynomial model of Bolton et al. (2012), here cast entirely in log-flux so that the multiplicative continuum becomes additive.

6 Adding the spectral resolution

A finite, lower instrumental resolution convolves the spectrum with a line-spread function, which we take to be Gaussian. At constant resolving power $R = \lambda/\Delta\lambda$, the LSF has a constant width *in velocity*, $\text{FWHM}_v = c/R$, hence a constant velocity dispersion

$$\sigma_v = \frac{c}{f R}, \quad f \equiv 2\sqrt{2 \ln 2}. \quad (22)$$

On the log-wavelength grid, which is uniform in velocity with $\text{velscale} = c \delta \text{ km s}^{-1}$ per pixel, this is a constant width in *pixels*,

$$\sigma_{\text{pix}} = \frac{\sigma_v}{\text{velscale}} = \frac{c}{f R \text{velscale}}. \quad (23)$$

A convolution by a Gaussian of fixed pixel width is a single multiply in the Fourier domain: a Gaussian of standard deviation σ_{pix} has transform $\exp(-2\pi^2\omega^2\sigma_{\text{pix}}^2)$, so we broaden each template “on the fly” by

$$\hat{T}_k(\omega) \longrightarrow \hat{T}_k(\omega) \exp(-2\pi^2\omega^2\sigma_{\text{pix}}^2). \quad (24)$$

The Fourier shift (14) and the Fourier Gaussian (24) compose trivially as a single elementwise product in the frequency domain,

$$\hat{T}_k(\omega) \longrightarrow \hat{T}_k(\omega) \exp(-2\pi^2\omega^2\sigma_{\text{pix}}^2) e^{-2\pi i\omega p}, \quad (25)$$

so the broadened, shifted template at any real (p, σ_{pix}) costs one inverse FFT. In particular the $\mathbf{b}_T$ and $\mathbf{M}_{TP}$ cross-correlations at fixed R are still FFTs — broaden the template transform, then correlate.

We promote R (equivalently σ_{pix}) to a *second nonlinear, outer* parameter alongside p . The structure is unchanged: at fixed (p, R) the model (16) is still linear in $\boldsymbol{\theta} = (\mathbf{w}, \mathbf{c})$, so the same separable scheme applies — a 2-D outer search/optimisation over (p, R) wrapping the inner linear solve (21).

An honest caveat: convolution and the logarithm do not commute. The LSF physically convolves the *linear* flux. But the inner solve is linear in $\boldsymbol{\theta}$ only if we broaden the *log-rectified* templates, which keeps the additive log-continuum intact. These are not the same operation, because convolution does not commute with the logarithm. Writing the rectified log-flux as $M(x)$ and the LSF as a normalised kernel, a cumulant (second-order) expansion of the exact linear-flux broadening gives

$$\ln(\text{LSF} * e^M) \approx \text{LSF} * M + \frac{1}{2} \text{Var}_{\text{LSF}}(M) + \dots, \quad (26)$$

where $\text{Var}_{\text{LSF}}(M)$ is the local variance of M under the kernel. The leading term $\text{LSF} * M$ is precisely our log-space broadening (24); the $\frac{1}{2} \text{Var}$ term is the first correction, which is positive and largest where M varies sharply — i.e. in deep, saturated line cores — so log-space broadening slightly *over-deepens* those cores at second order. The continuum, being smooth on the scale of the LSF, has negligible variance under the kernel and so factors out, $\text{LSF} * (\text{smooth}) \approx \text{smooth}$, and remains additive — exactly the property we need for linearity. We therefore conclude that the log-space convolution is a *first-order-accurate* approximation: excellent for shallow and moderate lines, with a small, bounded core bias for the deepest lines. The exact linear-flux convolution, by contrast, would mix the components nonlinearly (broadening $\sum_k w_k T_k$ in linear flux is not $\sum_k w_k$ times broadened T_k unless one broadens a *single* fixed spectrum), breaking the linearity of the inner solve for a multi-component basis.

Removing the core bias by iteration. The bias is, however, removable without giving up the linear inner solve, by a fixed-point iteration. Write the log-space-broadened model block as $A_c = \text{LSF} * M_c$ and the exact one as $E_c = \ln(\text{LSF} * e^{M_c})$, where M_c is the (un-broadened, shifted) log-flux of component $c \in \{\text{star}, \text{tell}\}$ from the current solution. Their difference $\delta = \sum_c (E_c - A_c)$ is precisely the $\frac{1}{2} \text{Var}$ -and-higher correction (concentrated in the cores). Since the exact model is $E = A + \delta$, fitting the *linear* machinery to the corrected data $\mathbf{Y} - \delta$ makes the exact model match the data; as δ depends on the solution we iterate $\delta \rightarrow \text{fit} \rightarrow \delta$ to a fixed point. Each pass is the usual linear solve (the normal matrix is unchanged; only the right-hand side moves) plus one forward linear-flux convolution, and the nonlinear parameters are re-optimised against the corrected data, so the resolution bias is removed as well. On an in-basis test the reduced χ^2 falls from ~ 40 to ~ 1 and the recovered R bias from $\sim -2\%$ to $\lesssim 0.1\%$; the iteration converges geometrically (a handful

to ~ 10 steps). This is the `n_conv_iter` option (off by default, which leaves the fast first-order approximation untouched).

Beyond a single Gaussian. For a wavelength-dependent $R(\lambda)$ or a non-Gaussian LSF there is no closed-form Fourier multiply. The practical remedy is to precompute banded (sparse) convolution matrices on a grid of R values and interpolate among them at fit time, retaining the linear inner solve at the cost of the single-multiply FFT shortcut.

7 Tellurics and the instrument wavelength zero-point

Telluric absorption from Earth’s atmosphere is imprinted at *fixed observed (topocentric) wavelengths*, independent of the star’s motion. We model it with a second rectified basis $\mathbf{S} = [S_0 \mid S_1 \mid \cdots \mid S_J]$, built exactly as the stellar templates (log-rectify + PCA) but from a grid of telluric transmission models. Because the tellurics do not share the stellar Doppler shift, $\mathbf{S}$ is *not* translated by Δx ; like the continuum it enters the design as shift-independent columns, broadened by the instrument LSF only (no rotational broadening). The forward model (16) becomes

$$Y(x) = \underbrace{\sum_k w_k T_k(x - \Delta x)}_{\text{stellar}} + \underbrace{\sum_j u_j S_j(x)}_{\text{telluric (observed frame)}} + \underbrace{\sum_m c_m P_m(x)}_{\text{continuum}}, \quad (27)$$

with the design augmented to $\mathbf{A}(p) = [\mathbf{T}(p) \mid \mathbf{S} \mid \mathbf{P}]$ and the linear vector to $\boldsymbol{\theta} = (\mathbf{w}, \mathbf{u}, \mathbf{c})$. The telluric and continuum blocks are shift-independent, so their normal-equation contributions are precomputed once; only the stellar auto- and cross-blocks depend on p , and the separable scan/solve is otherwise unchanged.

The wavelength zero-point. Real spectrographs carry a slit-dependent *wavelength zero-point* error (e.g. uncorrected flexure on Keck/DEIMOS): an overall shift of the observed wavelength scale by some lag p_0 . The stellar template then matches the data at $p_{\text{star}} = p_{\text{true}} + p_0$, conflating the true Doppler shift with the instrumental zero-point. Because the tellurics are at rest in the observer frame, their apparent shift measures p_0 *directly*. We therefore optionally promote a telluric lag p_{tell} to a further nonlinear parameter: the telluric columns are translated by a Fourier phase ramp, $\hat{S}_j(\omega) \rightarrow \hat{S}_j(\omega) e^{-2\pi i \omega p_{\text{tell}}}$, exactly as the stellar templates are translated by p (Section 3). At fixed (p, R, p_{tell}) the model is still linear in $(\mathbf{w}, \mathbf{u}, \mathbf{c})$, so the same separable least squares applies.

Unlike the radial velocity p — whose profiled χ^2 is multimodal and so requires the global FFT scan of Section 3 — the zero-point is small and its surface is smooth and unimodal near $p_{\text{tell}} = 0$. We therefore do *not* add a coarse grid dimension for it: the coarse scan keeps the tellurics at $p_{\text{tell}} = 0$ (in the shift-independent block) to locate the (p, R) basin, and p_{tell} is initialised at 0 and refined *jointly* with the other nonlinear parameters by the autodiff Newton step of Section 8.

Zero-point-corrected velocity. Since $p_{\text{star}} = p_{\text{true}} + p_0$ and $p_{\text{tell}} = p_0$, the instrumental zero-point cancels in the *difference*, giving the corrected radial velocity

$$\hat{v}_{\text{corr}} = c \left(e^{(p_{\text{star}} - p_{\text{tell}}) \delta} - 1 \right), \quad (28)$$

with its uncertainty propagated from the joint $(p_{\text{star}}, p_{\text{tell}})$ block of the nonlinear-parameter covariance. A barycentric correction is then added to place $\hat{v}_{\text{corr}}$ in the heliocentric frame. This reproduces the telluric/ sky-line wavelength recalibration used by DEIMOS pipelines (e.g. `dmmost`).

8 Optimisation and uncertainties

The fit proceeds in two stages, matched to the convexity structure of Section 4.

(1) Global coarse scan. The FFT machinery of Section 3 evaluates the profiled objective (21) on a dense grid of integer lags p across the velocity search window (and, when resolution is fit, on a coarse grid of R values via the broadening (24), using the cheap approximation $\mathbf{M}_{TT}(p) \approx \mathbf{M}_{TT}(0)$). The minimum over this 1-D (or 2-D) surface locates the basin of the dominant peak, immune to the sidelobes.

(2) Local refinement by autodiff Newton. Starting from the coarse minimiser, the profiled χ^2 as a function of the nonlinear parameters — p , or (p, σ_{pix}) when R is fit — is minimised locally. Because the radial-velocity translation (14), the Gaussian broadening (24), and the inner linear solve (21) are all differentiable operations, the gradient and Hessian of the profiled χ^2 are obtained *exactly* by JAX automatic differentiation. The refinement is a damped (Levenberg–Marquardt) Newton iteration: at each step the symmetrised Hessian is floored to positive definite by LM damping, the resulting step is forced to be a descent direction, and its length is set by a backtracking line search enforcing the Armijo sufficient-decrease condition. On the smooth, low-dimensional, locally near-parabolic main peak this converges in a few iterations.

Exact final solve. At the refined optimum $(\hat{p}, \hat{\sigma}_{\text{pix}})$ the template auto-block $\mathbf{M}_{TT}$ is recomputed *exactly* (no $p = 0$ approximation), the full design $\mathbf{A}(\hat{p})$ is assembled, and the linear system (21) is solved once more to deliver $\hat{\boldsymbol{\theta}} = (\hat{\mathbf{w}}, \hat{\mathbf{c}})$, the linear-parameter covariance $\mathbf{N}^{-1}$, and the exact χ^2 .

Uncertainties. The covariance of the nonlinear parameters follows from the curvature of the profiled objective at the optimum. For a least-squares χ^2 the standard $\Delta\chi^2 = 1$ rule gives

$$\text{Cov}(\hat{p}, \hat{\sigma}_{\text{pix}}) = 2 [\mathbf{H}]^{-1}, \quad \mathbf{H} = \nabla^2 \chi^2(\hat{p}, \hat{\sigma}_{\text{pix}}), \quad (29)$$

the factor 2 arising because $\frac{1}{2}\mathbf{H}$ is the Fisher information of the Gaussian likelihood $\propto \exp(-\chi^2/2)$. This covariance is propagated to the physical parameters (v, R) by the Jacobian of $v = c(e^{p\delta} - 1)$ and (23); e.g. $\partial v / \partial p = c\delta e^{p\delta}$ and $\partial R / \partial \sigma_{\text{pix}} = -R / \sigma_{\text{pix}}$. When the data are consistent with no broadening ($\hat{\sigma}_{\text{pix}} \rightarrow 0$), R is reported as a one-sided lower limit and only v is refined.

Degrees of freedom. The reduced chi-square uses

$$\text{dof} = n_{\text{good}} - (K + 1) - (J + 1) - (L + 1) - n_{\text{nonlin}}, \quad (30)$$

with n_{good} the number of unmasked pixels, $(K + 1)$ template weights, the $(J + 1)$ telluric weights when a telluric basis is used (otherwise drop this term), $(L + 1)$ continuum coefficients, and n_{nonlin} the number of fitted nonlinear parameters: the RV shift p , plus optionally the resolution σ_{pix} , the rotation $v \sin i$, and the telluric shift p_{tell} .

References

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Symbol	Type	Meaning
λ	–	wavelength
c	scalar	speed of light
$x = \ln \lambda$	–	log-wavelength coordinate
x_i	pixels	uniform log-wavelength grid, $i = 1, \dots, n$
δ	scalar	constant pixel spacing in x , $\delta = x_{i+1} - x_i$
n	scalar	number of pixels
v	scalar	radial velocity
$\Delta x = \ln(1 + v/c)$	scalar	log-wavelength shift induced by v
$p = \Delta x / \delta$	pixels	translation in pixels (the “lag”)
d_i	vector	observed (linear) flux in pixel i
σ_i	vector	1σ noise on d_i
$Y_i = \ln d_i$	vector	log-flux (the modelled quantity)
$C_i = (\sigma_i / d_i)^2$	vector	variance of Y_i ; weight $C_i^{-1} = 0$ if masked
$\mathbf{C} = \text{diag}(C_i)$	matrix	diagonal data covariance of $\mathbf{Y}$; weight $\mathbf{C}^{-1}$
$T_0 = \mu$	vector	mean rectified log-flux template
$T_k = \phi_k$	vector	k -th rectified-log-flux PCA component, $k = 1, \dots, K$
$\mathbf{T}$	$n \times (K+1)$	template design block, columns $[\mu, \phi_1, \dots, \phi_K]$
K	scalar	number of PCA components
$\mathbf{w} = (w_0, \dots, w_K)$	vector	template weights (w_0 multiplies μ)
P_m	vector	m -th Legendre continuum column, $m = 0, \dots, L$
$\mathbf{P}$	$n \times (L+1)$	continuum design block
L	scalar	continuum (Legendre) order
$\mathbf{c} = (c_0, \dots, c_L)$	vector	continuum coefficients
$S_j, \mathbf{S}$	vector, $n \times (J+1)$	telluric templates / block (observed frame, not RV-shifted)
$\mathbf{u} = (u_0, \dots, u_J)$	vector	telluric weights
p_{tell}	pixels	telluric lag (instrument wavelength zero-point)
$\mathbf{A}$	$n \times D$	full design $[\mathbf{T} \mid \mathbf{P}]$ (or $[\mathbf{T} \mid \mathbf{S} \mid \mathbf{P}]$ with tellurics)
$\boldsymbol{\theta} = (\mathbf{w}, \mathbf{c})$	vector	full linear parameter vector
$\mathbf{N} = \mathbf{A}^\top \mathbf{C}^{-1} \mathbf{A}$	matrix	normal matrix
$\mathbf{r} = \mathbf{A}^\top \mathbf{C}^{-1} \mathbf{Y}$	vector	right-hand side
$\mathbf{M}_{TT}, \mathbf{M}_{TP}, \mathbf{M}_{PP}$	matrix	blocks of $\mathbf{N}$
$\mathbf{b}_T, \mathbf{b}_P$	vector	blocks of $\mathbf{r}$
$R = \lambda / \Delta \lambda$	scalar	spectral resolving power
σ_v	scalar	LSF velocity dispersion
σ_{pix}	scalar	LSF width in pixels on the log grid
$\text{velscale} = c \delta$	scalar	velocity per pixel
$f = 2\sqrt{2 \ln 2}$	scalar	FWHM-to- σ conversion factor
ω	vector	Fourier frequency (cycles per pixel)
rfft, irfft	–	real forward / inverse FFT
χ^2	scalar	weighted least-squares objective
$\widehat{(\cdot)}$	–	estimate of a quantity

Table 1: Strict notation used throughout. Symbols are defined once here.